ShopEase – Test Plan

1. Project Overview

ShopEase is a fictional e-commerce application created as a Software Quality Assurance (SQA) portfolio project. The purpose of this project is to demonstrate manual testing skills, including requirement analysis, test case design, test documentation, and defect reporting.

This portfolio project currently focuses on testing the login functionality. The requirements and expected behaviours are defined as assumptions for learning purposes and have not yet been verified against a working application.

2. Testing Objectives

The objectives of this project are:

- To verify the expected behaviour of the ShopEase login functionality.
- To design clear and structured manual test cases based on defined requirements.
- To identify potential usability and validation issues during test execution.
- To document actual test results and report defects when they are observed.
- To maintain traceability between requirements and test cases.
- To demonstrate fundamental manual testing and QA documentation skills in an SQA portfolio.

3. Scope of Testing

3.1 In Scope

The following login-related areas are included in this portfolio project:

- Login using valid credentials.
- Login using an incorrect password.
- Login attempts with empty required fields.
- Validation of incorrectly formatted email addresses.
- Login attempts using an unregistered email address.
- Password visibility functionality, if available.
- Forgot Password navigation, if available.
- Sign Up navigation, if available.

3.2 Out of Scope

The following areas are outside the current testing scope:

- Payment processing and checkout.
- Shopping cart and product catalogue.
- Order placement and order tracking.
- Delivery and shipping functionality.
- Performance and load testing.
- Security penetration testing.
- Automated testing.

This scope is limited to login-related manual testing and may be updated if the project expands.

4. Testing Approach

The project will follow a manual, requirement-based testing approach.

4.1 Testing Techniques

- **Positive Testing:** Verify login behaviour using valid credentials.
- **Negative Testing:** Verify that invalid credentials and invalid inputs are handled appropriately.
- **Boundary and Input Validation Testing:** Check required fields and email-format validation.
- **Functional Testing:** Verify that login-related links and password visibility controls behave as expected, where available.
- **Requirements Traceability:** Link each test case to its corresponding requirement ID.

4.2 Test Execution Process

1. Review the defined requirements.
2. Prepare test cases and test data.
3. Execute each test case against a suitable working application or test environment.
4. Record the actual result for each test case.
5. Compare actual results with expected results.
6. Record a defect report if an actual result differs from the expected result.
7. Summarize the test execution results.

4.3 Current Project Status

The test cases and requirements are currently documented for a fictional e-commerce application. Test execution has not started because a working ShopEase application or mock environment has not yet been established.

5. Test Environment

5.1 Tools Used

- **Google Sheets:** Requirements, test cases, and bug report documentation.
- **Google Docs:** Test Plan documentation.
- **Web Browser:** Planned environment for testing the application's login interface.
- **GitHub:** Planned platform for version control and portfolio presentation.

5.2 Application Environment

ShopEase is currently a fictional e-commerce case study. A working application or mock login interface has not yet been configured.

Actual test execution will begin only after a suitable test environment is established. Any browser, device, or operating-system details will be recorded when they are confirmed during testing.

6. Entry and Exit Criteria

6.1 Entry Criteria

Testing may begin when:

- The requirements for the feature under test have been reviewed.
- Test cases have been prepared and reviewed.
- Suitable test data is available.
- A working application or mock test environment is accessible.
- The expected behaviour of the feature is defined.

6.2 Exit Criteria

The testing phase may be considered complete when:

- All planned test cases have been executed or their blocked status has been documented.
- Actual results have been recorded for executed test cases.
- Observed defects have been documented and tracked.
- Test execution results have been summarized.
- Any remaining limitations or untested scenarios have been clearly reported.

6.3 Current Status

The project has not met the entry criteria for test execution because a working application or mock environment is not yet available.

7. Test Deliverables

The following deliverables are planned for this project:

1. **Requirements Document:** Defines the assumed functional requirements for the ShopEase login feature.
2. **Test Cases:** Documents test scenarios, preconditions, steps, test data, expected results, and execution status.
3. **Bug Reports:** Provides a structured format for recording defects discovered during actual testing.
4. **Test Plan:** Describes the testing objectives, scope, approach, environment, and completion criteria.
5. **Test Execution Report:** Will summarize the number of tests executed, passed, failed, blocked, and not run.
6. **Project README:** Will explain the project purpose, contents, current status, and limitations for portfolio reviewers.

Deliverables will be updated as the project progresses. Execution results and defect reports will reflect actual observations rather than assumed outcomes.

8. Risks and Limitations

8.1 Risks

- The absence of a working application may delay test execution.
- Assumed requirements may differ from the behaviour of a real e-commerce application.
- Test results may be incomplete until a suitable test environment is available.
- Some features, such as password visibility and password recovery, may not be implemented in the eventual test environment.

8.2 Limitations

- This project currently focuses on login-related manual testing.
- The requirements are defined for educational and portfolio purposes.
- No test cases have been executed against a working application.
- No defects have been confirmed or reported.
- Automation, performance testing, and security testing are not included in the current scope.

8.3 Future Improvements

- Establish a working ShopEase test environment.
- Execute the documented test cases and record actual results.
- Document and track any confirmed defects.
- Expand coverage to registration, product search, shopping cart, and checkout.
- Publish the completed project documentation on GitHub.